Engineering Economy

Chapter 6:

Comparison and Selection Among Alternatives

The objective of chapter 6 is to evaluate correctly capital investment alternatives when the time value of money is a key influence.

Chapter 6
6.1. Introduction

- Most engineering projects can be accomblished by more than one feasible design alternative.
- Typically, the alternatives being considered require the investment of amounts of capital, and their annual revenues and cost may vary.

Making decisions means comparing alternatives.

- In this chapter we examine *feasible design alternatives*.
- The decisions considered are those selecting from among a set of *mutually exclusive* alternatives—when selecting one excludes the choice of any of the others.

Mutually exclusive alternatives (MEAs)

- We examine these on the basis of economic considerations alone.
- The alternatives may have different initial investments and their annual revenues and costs may vary.
- The alternatives must provide comparable “usefulness”: performance, quality, etc.
- The basic methods from chapter 5 provide the basis for economic comparison of the alternatives.

Apply this rule, based on Principle 2 from Chapter 1.

The alternative that requires the minimum investment of capital and produces satisfactory functional results will be chosen unless the incremental capital associated with an alternative having a larger investment can be justified with respect to its incremental benefits. This alternative is the *base alternative*.

For alternatives that have a larger investment than the base…

If the extra benefits obtained by investing additional capital are better than those that could be obtained from investment of the same capital elsewhere in the company at the MARR, the investment should be made.

(Please note that there are some cautions when considering more than two alternatives, which will be examined later.)

Chapter 6
6.2. Basic Concepts for Comparing Alternatives

6.2.1. Investment and Cost Alternatives :

There are two basic types of alternatives

Investment Alternatives

Those with initial (or front-end) capital investment that produces positive cash flows from increased revenue, savings through reduced costs, or both.

Cost Alternatives

Those with all negative cash flows, except for a possible positive cash flow from disposal of assets at the end of the project's useful life.

Select the alternative that gives you the most money!

- For *investment alternatives* the PW of all cash flows must be positive, at the MARR, to be attractive. Select the alternative with the largest PW.
- For *cost alternatives* the PW of all cash flows will be negative. Select the alternative with the largest (smallest in absolute value) PW.

Chapter 6
6.2. Basic Concepts for Comparing Alternatives

Investment alternative example :

Use a MARR of 10% and useful life of 5 years to select between the investment alternatives below.

	Alternative	
	A	B
Capital investment	-$100,000	-$125,000
Annual revenues less expenses	$34,000	$41,000

PW_A = - 100,000 + 34,000(P/A, 10 %, 5) = 28,887
PW_B = - 125,000 + 41,000(P/A, 10 %, 5) = 30,423

Both alternatives are attractive, but Alternative B provides a greater present worth, so is better economically.

Chapter 6
6.2. Basic Concepts for Comparing Alternatives

Cost alternative example

Use a MARR of 12% and useful life of 4 years to select between the cost alternatives below.

	Alternative	
	A	B
Capital investment	-$80,000	-$60,000
Annual expenses	-$25,000	-$30,000

PW_A = - 80,000 - 25,000(P/A, 12 %, 4) = - 155,933
PW_B = - 60,000 - 30,000(P/A, 12 %, 4) = - 151,119

Alternative B costs less than Alternative A, it has a greater PW, so is better economically.

Pause and solve

Your local foundry is adding a new furnace. There are several different styles and types of furnaces, so the foundry must select from among a set of mutually exclusive alternatives. Initial capital investment and annual expenses for each alternative are given in the table below. None have any market value at the end of its useful life. Using a MARR of 15%, which furnace should be chosen?

	Furnace		
	F1	F2	F3
Investment	$110,000	$125,000	$138,000
Useful life	10 years	10 years	10 years
Total annual expenses	$53,800	$51,625	$45,033

Solution

Using a MARR of 15%, the PW is shown for each of the three alternatives in the table below.

	Furnace		
	F1	F2	F3
Investment	$110,000	$125,000	$138,000
Useful life	10 years	10 years	10 years
Total annual expenses	$53,800	$51,625	$45,033
Present Worth @ 15%	-$380,010	-$384,094	-$364,010

The largest value is -$364,010, indicating that Furnace F3 is the best alternative.

Chapter 6
6.2. Basic Concepts for Comparing Alternatives

Remember ! In section 2 two rules were given;

Rule 1: When revenues and other economic benefits are present and vary among the alternatives, choose the alternative that maximizes overall profitability.

That is, select the alternative that has **the greatest positive equivalent worth** at i=MARR

Rule 2: When revenues and other economic benefits are not present or one constant among the alternatives, consider only the cost and select the alternative that minimizes total cost

That is, select the alternative that has **the least negative equivalent worth** at i=MARR

Chapter 6
6.3. The Study (Analysis) Period

Determining the study period.

- A *study period* (or *planning horizon*) is the time period over which MEAs are compared, and it must be appropriate for the decision situation.
- MEAs can have *equal* lives (in which case the study period used is these equal lives), or they can have *unequal* lives, and at least one does not match the study period.
- The equal life case is straightforward, and was used in the previous two examples.

The determination of the study period for a decision situation may be influenced by several factors, such as;

- The service period required
- The useful life of the shorter-lived alternative
- The useful life of the longer-lived alternative
- Company policies

Chapter 6
6.3. The Study (Analysis) Period

The useful life of an asset is the period during which it is kept in productive use in a trade or business.

The useful lives of alternatives being compared, relative to the selected study period, can involve two situations:

1- Useful lives are the same for all alternatives and equal to the study period.

2- Useful lives unequal among the alternatives, and at least one does not match the study period.

Chapter 6
6.4. Useful Lives Equal to the Study Period

Useful lives are the same for all alternatives and equal to the study period.

In this part we will discuss the comparison of mutually exclusive alternatives, using;

- Equivalent-worth methods

- Rate- of- return methods

Chapter 6
6.4. Useful Lives Equal to the Study Period

6.4.1 Equivalent – worth methods

In chapter 5, we learned that equivalent-worth methods convert all relevant cash flows into equivalent present, annual, or future amounts.

The economic ranking of alternatives will be the same when using the three methods.

Consider the general case of two alternatives A and B.

If $PW\ (i\%)_A < PW\ (i\%)_B$, then the FW and AW analyses will result in the same preference for alternative B.

Chapter 6
6.4. Useful Lives Equal to the Study Period

EXAMPLE 6-1 **Analyzing Investment Alternatives by Using Equivalent Worth**

Best Flight, Inc., is considering three mutually exclusive alternatives for implementing an automated passenger check-in counter at its hub airport. Each alternative meets the same service requirements, but differences in capital investment amounts and benefits (cost savings) exist among them. The study period is 10 years, and the useful lives of all three alternatives are also 10 years. Market values of all alternatives are assumed to be zero at the end of their useful lives. If the airline's MARR is 10% per year, which alternative should be selected in view of the cash-flow diagrams shown on page 248?

Solution by the PW Method

$$PW(10\%)_A = -\$390{,}000 + \$69{,}000(P/A, 10\%, 10) = \$33{,}977,$$

$$PW(10\%)_B = -\$920{,}000 + \$167{,}000(P/A, 10\%, 10) = \$106{,}148,$$

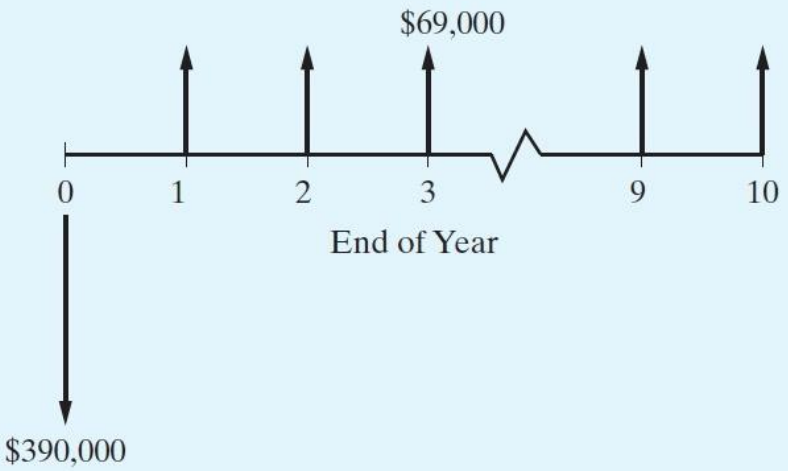
$$PW(10\%)_C = -\$660{,}000 + \$133{,}500(P/A, 10\%, 10) = \$160{,}304.$$

Based on the PW method, Alternative C would be selected because it has the largest PW value ($160,304). The order of preference is $C \succ B \succ A$, where $C \succ B$ means C is preferred to B.

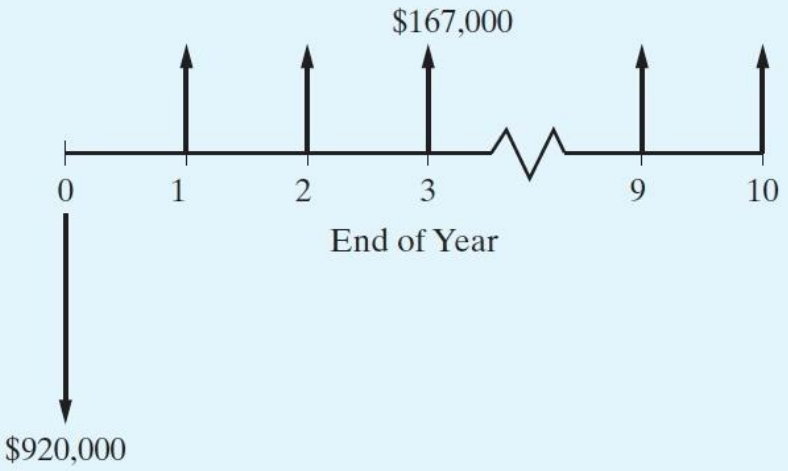
Chapter 6
6.4. Useful Lives Equal to the Study Period

EXAMPLE 6-1 **Analyzing Investment Alternatives by Using Equivalent Worth**

Alternative *A*:



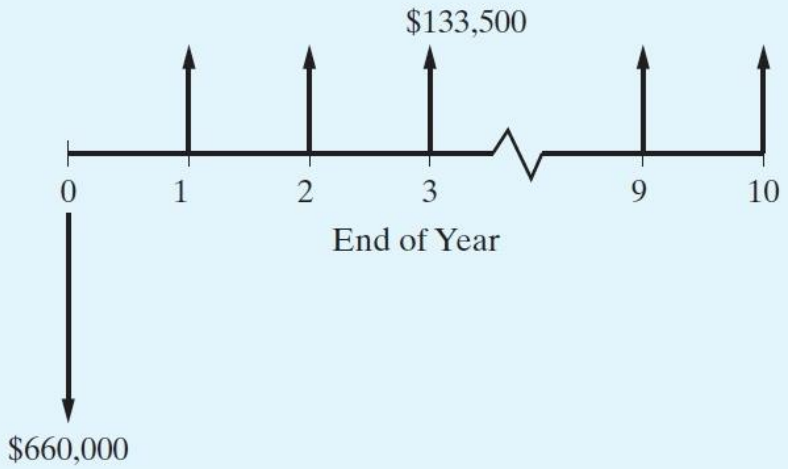
Alternative *B*:



Chapter 6
6.4. Useful Lives Equal to the Study Period

EXAMPLE 6-1 **Analyzing Investment Alternatives by Using Equivalent Worth**

Alternative *C*:



Solution by the AW Method

$$AW(10\%)_A = -\$390{,}000(A/P, 10\%, 10) + \$69{,}000 = \$5{,}547,$$

$$AW(10\%)_B = -\$920{,}000(A/P, 10\%, 10) + \$167{,}000 = \$17{,}316,$$

$$AW(10\%)_C = -\$660{,}000(A/P, 10\%, 10) + \$133{,}500 = \$26{,}118.$$

Alternative *C* is again chosen because it has the largest AW value ($26,118).

Chapter 6
6.4. Useful Lives Equal to the Study Period

EXAMPLE 6-1 **Analyzing Investment Alternatives by Using Equivalent Worth**

Solution by the FW Method

$$FW(10\%)_A = -\$390{,}000(F/P, 10\%, 10) + \$69{,}000(F/A, 10\%, 10) = \$88{,}138,$$

$$FW(10\%)_B = -\$920{,}000(F/P, 10\%, 10) + \$167{,}000(F/A, 10\%, 10) = \$275{,}342,$$

$$FW(10\%)_C = -\$660{,}000(F/P, 10\%, 10) + \$133{,}500(F/A, 10\%, 10) = \$415{,}801.$$

Based on the FW method, the choice is again Alternative C because it has the largest FW value ($415,801). For all three methods (PW, AW, and FW) in this example, notice that $C \succ B \succ A$ because of the equivalency relationship among the methods. Also, notice that Rule 1 (Section 6.2.2) applies in this example, since the economic benefits (cost savings) vary among the alternatives.

Chapter 6
6.4. Useful Lives Equal to the Study Period

EXAMPLE 6-2 **Analyzing Cost-Only Alternatives, Using Equivalent Worth**



A company is planning to install a new automated plastic-molding press. Four different presses are available. The initial capital investments and annual expenses for these four mutually exclusive alternatives are as follows:

	Press			
	P1	*P2*	*P3*	*P4*
Capital investment	$24,000	$30,400	$49,600	$52,000
Useful life (years)	5	5	5	5
Annual expenses				
Power	2,720	2,720	4,800	5,040
Labor	26,400	24,000	16,800	14,800
Maintenance	1,600	1,800	2,600	2,000
Property taxes and insurance	480	608	992	1,040
Total annual expenses	$31,200	$29,128	$25,192	$22,880

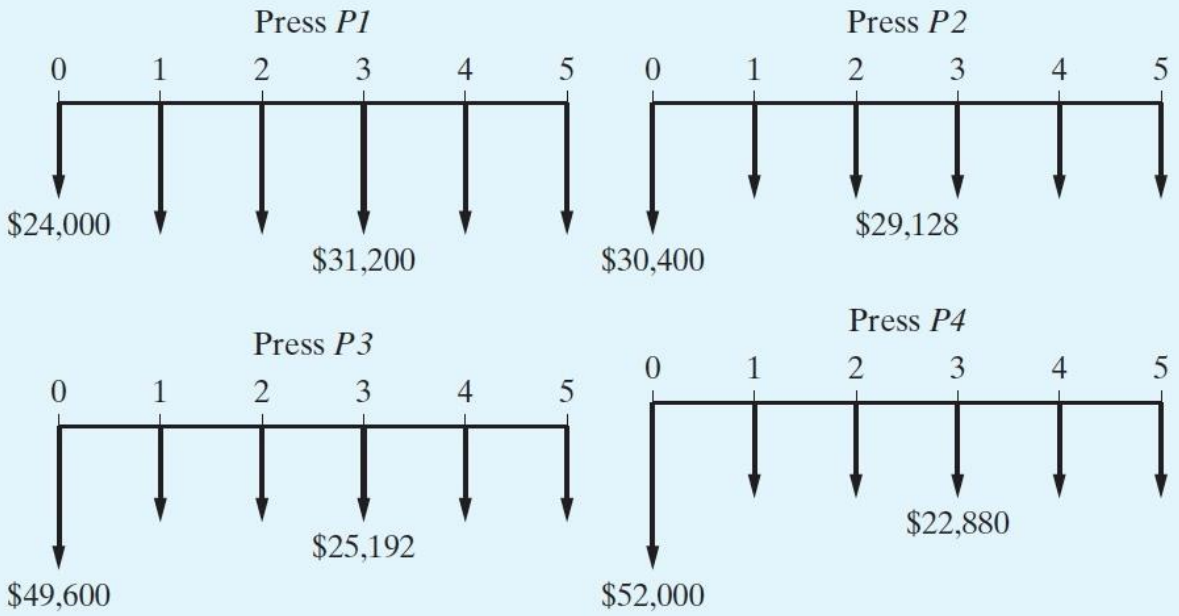
Assume that each press has the same output capacity (120,000 units per year) and has no market value at the end of its useful life; the selected analysis period is five years; and any additional capital invested is expected to earn at least 10% per year. Which press should be chosen if 120,000 nondefective units per year are produced by each press and all units can be sold? The selling price is $0.375 per unit. Solve by hand and by spreadsheet.

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6.4. Useful Lives Equal to the Study Period

EXAMPLE 6-2 **Analyzing Cost-Only Alternatives, Using Equivalent Worth**

Solution by Hand

Since the same number of nondefective units per year will be produced and sold using each press, revenue can be disregarded (Principle 2, Chapter 1). The end-of-year cash-flow diagrams of the four presses are:



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6.4. Useful Lives Equal to the Study Period

EXAMPLE 6-2 **Analyzing Cost-Only Alternatives, Using Equivalent Worth**

The preferred alternative will minimize the equivalent worth of total costs over the five-year analysis period (Rule 2, page 245). That is, the four alternatives can be compared as cost alternatives. The PW, AW, and FW calculations for Alternative *P1* are

$$\begin{aligned}
\text{PW}(10\%)_{P1} &= -\$24{,}000 - \$31{,}200(P/A, 10\%, 5) = -\$142{,}273, \\
\text{AW}(10\%)_{P1} &= -\$24{,}000(A/P, 10\%, 5) - \$31{,}200 = -\$37{,}531, \\
\text{FW}(10\%)_{P1} &= -\$24{,}000(F/P, 10\%, 5) - \$31{,}200(F/A, 10\%, 5) \\
&= -\$229{,}131.
\end{aligned}$$

The PW, AW, and FW values for Alternatives *P2*, *P3*, and *P4* are determined with similar calculations and shown for all four presses in Table 6-1. Alternative *P4* minimizes all three equivalent-worth values of total costs and is the preferred alternative. The preference ranking ($P4 \succ P2 \succ P1 \succ P3$) resulting from the analysis is the same for all three methods.

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6.4. Useful Lives Equal to the Study Period

EXAMPLE 6-3 **Analyzing Alternatives with Different Reject Rates**

Consider the four plastic molding presses of Example 6-2. Suppose that each press is still capable of producing 120,000 total units per year, but the estimated reject rate is different for each alternative. This means that the expected revenue will differ among the alternatives since only nondefective units can be sold. The data for the four presses are summarized below. The life of each press (and the study period) is five years.

	Press			
	P1	*P2*	*P3*	*P4*
Capital investment	$24,000	$30,400	$49,600	$52,000
Total annual expenses	$31,200	$29,128	$25,192	$22,880
Reject rate	8.4%	0.3%	2.6%	5.6%

If all nondefective units can be sold for $0.375 per unit, which press should be chosen? Solve by hand and by spreadsheet.

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6.4. Useful Lives Equal to the Study Period

TABLE 6-1 Comparison of Four Molding Presses, Using the PW, AW, and FW Methods to Minimize Total Costs

	Press (Equivalent-Worth Values)			
Method	*P1*	*P2*	*P3*	*P4*
Present worth	−$142,273	−$140,818	−$145,098	−$138,734
Annual worth	−37,531	−37,148	−38,276	−36,598
Future worth	−229,131	−226,788	−233,689	−223,431

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6.4. Useful Lives Equal to the Study Period

EXAMPLE 6-3 **Analyzing Alternatives with Different Reject Rates**



Solution by Hand

In this example, each of the four alternative presses produces 120,000 units per year, but they have different estimated reject rates. Therefore, the number of nondefective output units produced and sold per year, as well as the annual revenues received by the company, varies among the alternatives. But the annual expenses are assumed to be unaffected by the reject rates. In this situation, the preferred alternative will maximize overall profitability (Rule 1, Section 6.2.2). That is, the four presses need to be compared as investment alternatives. The PW, AW, and FW calculations for Alternative *P4* are given below:

$$\begin{aligned}
PW(10\%)_{P4} &= -\$52{,}000 + [(1-0.056)(120{,}000)(\$0.375) - \$22{,}880](P/A, 10\%, 5) \\
&= \$22{,}300, \\
AW(10\%)_{P4} &= -\$52{,}000(A/P, 10\%, 5) + [(1 - 0.056)(120{,}000)(\$0.375) - \$22{,}880] \\
&= \$5{,}882, \\
FW(10\%)_{P4} &= -\$52{,}000(F/P, 10\%, 5) \\
&\quad + [(1 - 0.056)(120{,}000)(\$0.375) - \$22{,}800](F/A, 10\%, 5) \\
&= \$35{,}914.
\end{aligned}$$

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6.4. Useful Lives Equal to the Study Period

TABLE 6-2 Comparison of Four Molding Presses, Using the PW, AW, and FW Methods to Maximize Overall Profitability

	Press (Equivalent-Worth Values)			
Method	*P1*	*P2*	*P3*	*P4*
Present worth	$13,984	$29,256	$21,053	$22,300
Annual worth	3,689	7,718	5,554	5,882
Future worth	22,521	47,117	33,906	35,914

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Chapter 6
6.4. Useful Lives Equal to the Study Period

EXAMPLE 6-3 **Analyzing Alternatives with Different Reject Rates**

The PW, AW, and FW values for Alternatives *P1*, *P2*, and *P3* are determined with similar calculations and shown for all four alternatives in Table 6-2. Alternative *P2* maximizes all three equivalent-worth measures of overall profitability and is preferred [versus *P4* in Example 6-2]. The preference ranking ($P2 \succ P4 \succ P3 \succ P1$) is the same for the three methods but is different from the ranking in Example 6-2. The different preferred alternative and preference ranking are the result of the varying capability among the presses to produce nondefective output units.

Chapter 6
6.4. Useful Lives Equal to the Study Period

Comparing MEAs with equal lives.

When lives are equal adjustments to cash flows are not required. The MEAs can be compared by directly comparing their *equivalent worth* (*PW, FW, or AW*) calculated using the MARR. The decision will be the same regardless of the equivalent worth method you use. For a MARR of 12%, select from among the MEAs below.

	Alternatives			
	A	B	C	D
Capital investment	-$150,000	-$85,000	-$75,000	-$120,000
Annual revenues	$28,000	$16,000	$15,000	$22,000
Annual expenses	-$1,000	-$550	-$500	-$700
Market Value (EOL)	$20,000	$10,000	$6,000	$11,000
Life (years)	10	10	10	10

Selecting the best alternative.

Present worth analysis → select Alternative A (but C is close).

$$PW_A = -150,000 + 27,000(P/A, 12\%, 10) + 20,000(P/F, 12\%, 10) = 8{,}995$$

$$PW_B = -85,000 + 15,450(P/A, 12\%, 10) + 10,000(P/F, 12\%, 10) = 5{,}516$$

$$PW_C = -75,000 + 14,500(P/A, 12\%, 10) + 6,000(P/F, 12\%, 10) = 8{,}860$$

$$PW_D = -120,000 + 21,300(P/A, 12\%, 10) + 11,000(P/F, 12\%, 10) = 3{,}891$$

Annual worth analysis—the decision is the same.

$$AW_A = \$1,592 \qquad AW_C = \$1,568$$

$$AW_B = \$976 \qquad AW_D = \$689$$

Chapter 6
6.5. Useful Lives are Unequal among the Alternatives

Comparing MEAs with unequal lives.

- The repeatability assumption, when applicable, simplified comparison of alternatives.

- If repeatability cannot be used, an appropriate study period must be selected (the coterminated assumption). This is most often used in engineering practice because product life cycles are becoming shorter.

Chapter 6
6.5. Useful Lives are Unequal among the Alternatives

The useful life of an alternative is less than the study period.

- Cost alternatives
  - Contracting or leasing for remaining years may be appropriate
  - Repeat part of the useful life and use an estimated market value to truncate
- Investment alternatives
  - Cash flows reinvested at the MARR at the end of the study period
  - Replace with another asset, with possibly different cash flows, after the study period

The useful life of an alternative is greater than the study period.

- Truncate the alternative at the end of the study period, using an estimated market value.
- The underlying principle in all such analysis is to compare the MEAs in a decision situation over the same study (analysis) period.

Chapter 6
6.5. Useful Lives are Unequal among the Alternatives

6.5.1. Equivalent worth methods can be used for MEAs with unequal lives.

- If repeatability can be assumed, the MEAs are most easily compared by finding the annual worth (AW) of each alternative over its own useful life, and recommending the one having the most economical value.

- For cotermination, use any equivalent worth method using the cash flows available for the study period.

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6.5. Useful Lives are Unequal among the Alternatives

EXAMPLE 6-7 **Useful Lives ≠ Study Period: The Repeatability Assumption**

VIDEO SOLUTIONS

The following data have been estimated for two mutually exclusive investment alternatives, *A* and *B*, associated with a small engineering project for which revenues as well as expenses are involved. They have useful lives of four and six years, respectively. If MARR = 10% per year, show which alternative is more desirable by using equivalent-worth methods (computed by hand and by spreadsheet). Use the repeatability assumption.

	A	*B*
Capital investment	$3,500	$5,000
Annual cash flow	1,255	1,480
Useful life (years)	4	6
Market value at end of useful life	0	0

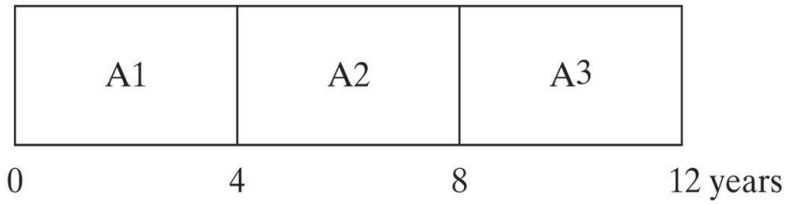
Solution

The least common multiple of the useful lives of Alternatives *A* and *B* is 12 years. Using the repeatability assumption and a 12-year study period, the first like (identical) replacement of Alternative *A* would occur at EOY four, and the second would be at EOY eight. For Alternative *B*, one like replacement would occur at EOY six. This is illustrated in Part 1 of Figure 6-9.

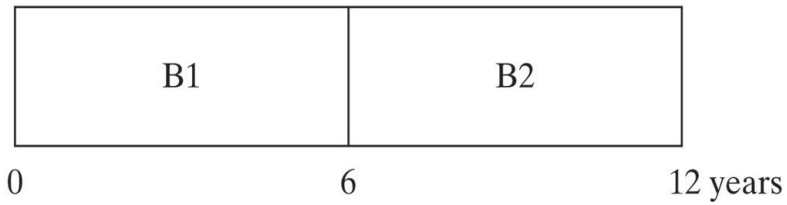
Chapter 6
6.5. Useful Lives are Unequal among the Alternatives

Part 1: Repeatability Assumption, Example 6-7, Least Common Multiple of Useful Lives Is 12 years.

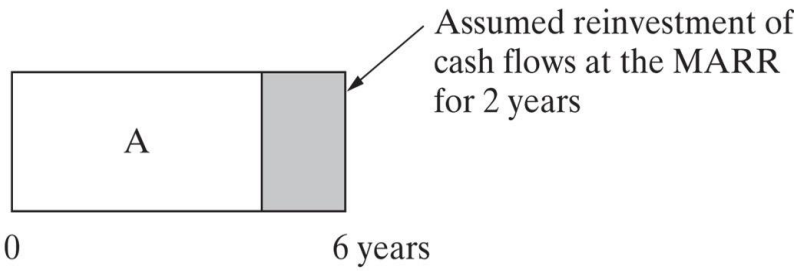
Three cycles of Alternative *A*:

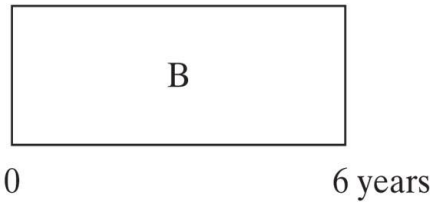


Two cycles of Alternative *B*:



Part 2: Coterminated Assumption, Example 6-8, Six-Year Analysis Period.





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Chapter 6
6.5. Useful Lives are Unequal among the Alternatives

EXAMPLE 6-7 **Useful Lives ≠ Study Period: The Repeatability Assumption**

Solution by the PW Method

The PW (or FW) solution must be based on the total study period (12 years). The PW of the initial useful life cycle will be different than the PW of subsequent replacement cycles:

$$\begin{aligned} PW(10\%)_A &= -\$3{,}500 - \$3{,}500[(P/F, 10\%, 4) + (P/F, 10\%, 8)] \\ &\quad + (\$1{,}255)(P/A, 10\%, 12) \\ &= \$1{,}028, \end{aligned}$$

$$\begin{aligned} PW(10\%)_B &= -\$5{,}000 - \$5{,}000(P/F, 10\%, 6) \\ &\quad + (\$1{,}480)(P/A, 10\%, 12) \\ &= \$2{,}262. \end{aligned}$$

Based on the PW method, we would select Alternative *B*.

Chapter 6
6.5. Useful Lives are Unequal among the Alternatives

EXAMPLE 6-7 **Useful Lives ≠ Study Period: The Repeatability Assumption**

Solution by the AW Method

The like replacement of assets assumes that the economic estimates for the initial useful life cycle will be repeated in each subsequent replacement cycle. Consequently, the AW will have the same value for each cycle and for the study period (12 years). This is demonstrated in the next AW solution by calculating (1) the AW of each alternative over the 12-year analysis period based on the previous PW values and (2) determining the AW of each alternative over one useful life cycle. Based on the previously calculated PW values, the AW values are

$$\text{AW}(10\%)_A = \$1{,}028(A/P, 10\%, 12) = \$151,$$
$$\text{AW}(10\%)_B = \$2{,}262(A/P, 10\%, 12) = \$332.$$

Next, the AW of each alternative is calculated over one useful life cycle:

$$\text{AW}(10\%)_A = -\$3{,}500(A/P, 10\%, 4) + (\$1{,}255) = \$151,$$
$$\text{AW}(10\%)_B = -\$5{,}000(A/P, 10\%, 6) + (\$1{,}480) = \$332.$$

This confirms that both calculations for each alternative result in the same AW value, and we would again select Alternative *B* because it has the larger value ($332).

Spreadsheet Solution

Figure 6-10 shows the spreadsheet solution for this example. EOY cash flows are computed for each alternative over the entire 12-year study period. For Alternative *A*, the annual cash flow of $1,255 is combined with the necessary

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6.5. Useful Lives are Unequal among the Alternatives

EXAMPLE 6-8 **Useful Lives ≠ Study Period: The Coterminated Assumption**

Suppose that Example 6-7 is modified such that an analysis period of six years is used (coterminated assumption) instead of 12 years, which was based on repeatability and the least common multiple of the useful lives. Perhaps the responsible manager did not agree with the repeatability assumption and wanted a six-year analysis period because it is the planning horizon used in the company for small investment projects.

Solution

An assumption used for an investment alternative (when useful life is less than the study period) is that all cash flows will be reinvested by the firm at the MARR until the end of the study period. This assumption applies to Alternative *A*, which has a four-year useful life (two years less than the study period), and it is illustrated in Part 2 of Figure 6-9. We use the FW method to analyze this situation:

$$\begin{aligned} FW(10\%)_A &= [-\$3{,}500(F/P,10\%,4) + (\$1{,}255)(F/A,10\%,4)](F/P,10\%,2) \\ &= \$847, \\ FW(10\%)_B &= -\$5{,}000(F/P,10\%,6) + (\$1{,}480)(F/A,10\%,6) \\ &= \$2{,}561. \end{aligned}$$

Based on the FW of each alternative at the end of the six-year study period, we would select Alternative *B* because it has the larger value ($2,561).

Chapter 6
6.5. Useful Lives are Unequal among the Alternatives

EXAMPLE 6-10 **AW and Repeatability: Perfect Together!**



Three products will be manufactured in a new facility at the Apex Manufacturing Company. They each require an identical manufacturing operation, but different production times, on a broaching machine. Two alternative types of broaching machines (*M1* and *M2*) are being considered for purchase. One machine type must be selected.

For the same level of annual demand for the three products, *annual* production requirements (machine hours) and annual operating expenses (per machine) are listed next. Which machine should be selected if the MARR is 20% per year? Solve by hand and by spreadsheet. Show all work to support your recommendation. (Use Rule 2 on page 245 to make your recommendation.)

Product	Machine *M1*	Machine *M2*
ABC	1,500 hr	900 hr
MNQ	1,750 hr	1,000 hr
STV	2,600 hr	2,300 hr
	5,850 hr	4,200 hr
Capital investment	$15,000 per machine	$22,000 per machine
Expected life	five years	eight years
Annual expenses	$4,000 per machine	$6,000 per machine

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EXAMPLE 6-10 **AW and Repeatability: Perfect Together!**

Assumptions: The facility will operate 2,000 hours per year. Machine availability is 90% for Machine *M1* and 80% for Machine *M2*. The yield of Machine *M1* is 95%, and the yield of Machine *M2* is 90%. Annual operating expenses are based on an assumed operation of 2,000 hours per year, and workers are paid during any idle time of Machine *M1* or Machine *M2*. Market values of both machines are negligible.



Solution by Hand

The company will need 5,850 hours/[2,000 hours (0.90)(0.95)] = 3.42 (four machines of type *M1*) or 4,200 hours/[2,000 hours (0.80)(0.90)] = 2.92 (three machines of type *M2*). The maximum operation time of 2,000 hours per year

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EXAMPLE 6-10 **AW and Repeatability: Perfect Together!**

in the denominator must be multiplied by the availability of each machine and the yield of each machine, as indicated.

The annual cost of ownership, assuming a MARR = 20% per year, is \$15,000(4)(*A*/*P*, 20%, 5) = \$20,064 for Machine *M1* and \$22,000(3)(*A*/*P*, 20%, 8) = \$17,200 for Machine *M2*.

There is an excess capacity when four Machine *M1*s and three Machine *M2*s are used to provide the machine-hours (5,850 and 4,200, respectively) just given. If we assume that the operator is paid for idle time he or she may experience on *M1* or *M2*, the annual expense for the operation of four M1s is 4 machines × \$4,000 per machine = \$16,000. For three *M2*s, the annual expense is 3 machines × \$6,000 per machine = \$18,000.

The total equivalent annual cost for four Machine *M1*s is \$20,064 + \$16,000 = \$36,064. Similarly, the total equivalent annual expense for three Machine *M2*s is \$17,200 + \$18,000 = \$35,200. By a slim margin, Machine *M2* is the preferred choice to minimize equivalent annual costs with the repeatability assumption.

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EXAMPLE 6-11 **Modeling Estimated Expenses as Arithmetic Gradients**

You are a member of an engineering project team that is designing a new processing facility. Your present design task involves the portion of the catalytic system that requires pumping a hydrocarbon slurry that is corrosive and contains abrasive particles. For final analysis and comparison, you have selected two fully lined slurry pump units, of equal output capacity, from different manufacturers. Each unit has a large-diameter impeller required and an integrated electric motor with solid-state controls. Both units will provide the same level of service (support) to the catalytic system but have different useful lives and costs.

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EXAMPLE 6-11 **Modeling Estimated Expenses as Arithmetic Gradients**

	Pump Model	
	SP240	HEPS9
Capital investment	$33,200	$47,600
Annual expenses:		
Electrical energy	$2,165	$1,720
Maintenance	$1,100 in year 1, and increasing $500/yr thereafter	$500 in year 4, and increasing $100/yr thereafter
Useful life (years)	5	9
Market value (end of useful life)	0	5,000

The new processing facility is needed by your firm at least as far into the future as the strategic plan forecasts operating requirements. The MARR is 20% per year. Based on this information, which slurry pump should you select?

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EXAMPLE 6-11 **Modeling Estimated Expenses as Arithmetic Gradients**

Solution

Notice that the estimates for maintenance expenses involve an arithmetic gradient series (Chapter 4). A cash-flow diagram is very useful in this situation to help keep track of the various cash-flow series. The cash-flow diagrams for pump models SP240 and HEPS9 are shown in Figure 6-12.

The repeatability assumption is a logical choice for this analysis, and a study period of either infinite or 45 years (least common multiple of the useful lives) in length can be used. With repeatability, the AW over the initial *useful life* of each alternative is the same as its AW over the length of either study period:

$$\begin{aligned}
AW(20\%)_{SP240} &= -\$33{,}200(A/P, 20\%, 5) - \$2{,}165 \\
&\quad -[\$1{,}100 + \$500(A/G, 20\%, 5)] \\
&= -\$15{,}187, \\
AW(20\%)_{HEPS9} &= -\$47{,}600(A/P, 20\%, 9) + \$5{,}000(A/F, 20\%, 9) \\
&\quad -\$1{,}720 - [\$500(P/A, 20\%, 6) \\
&\quad +\$100(P/G, 20\%, 6)] \times (P/F, 20\%, 3) \times (A/P, 20\%, 9) \\
&= -\$13{,}622.
\end{aligned}$$

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EXAMPLE 6-11 **Modeling Estimated Expenses as Arithmetic Gradients**

Based on Rule 2 (Section 6.2.2), you should select pump model HEPS9, since the AW over its useful life (nine years) has the smaller negative value (−$13,622).

As additional information, the following two points support in choosing the repeatability assumption in Example 6-11:

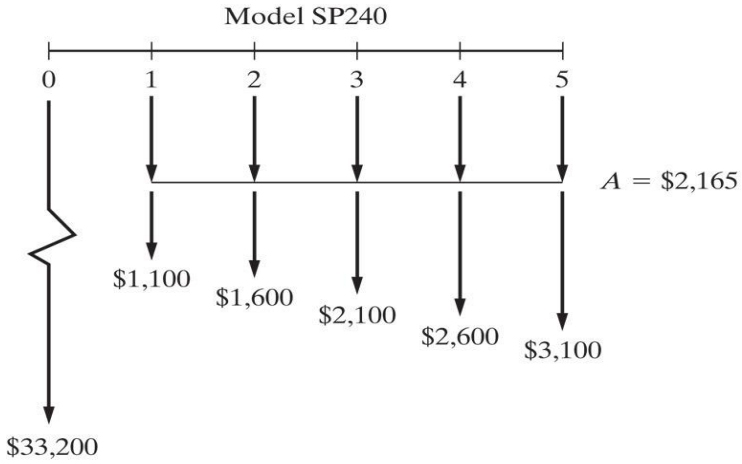
1. The repeatability assumption is commensurate with the long planning horizon for the new processing facility and with the design and operating requirements of the catalytic system.

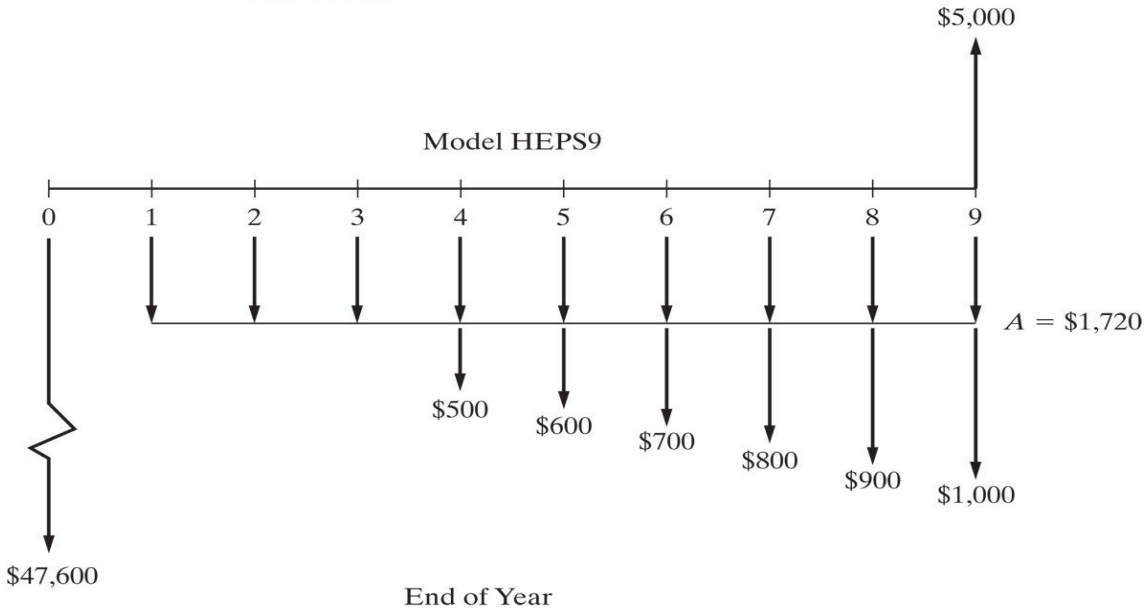
EXAMPLE 6-11 **Modeling Estimated Expenses as Arithmetic Gradients**

2. If the initial estimated costs change for future pump-replacement cycles, a logical assumption is that the ratio of the AW values for the two alternatives will remain approximately the same. Competition between the two manufacturers should cause this to happen. Hence, the pump selected (model HEPS9) should continue to be the preferred alternative.

If the existing model is redesigned or new models of slurry pumps become available, however, another study to analyze and compare all feasible alternatives is required before a replacement of the selected pump occurs.

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